An Indonesian limited-data ASR corpus for fair nine-architecture benchmarking — corpus statistics, balance verification, and split integrity (v7)

Article type: *Data in Brief* data article (template-compliant). **Generated**: 2026-05-30. **Source data**: `metadata/dataset_metadata_clean.csv` (102,544 rows × 23 columns) and the frozen `splits/{train,dev,test}_clean.tsv` splits (71,792 / 15,376 / 15,376), `dataset_version = v7_natural_synth`. These are the **exact files the nine paper models train and evaluate on**. Audio-quality figures are computed from a stratified random sample of n = 297 files (≈ 27 per category). Machine-readable summary in `stats/dataset_stats.json`; per-stratum breakdowns in `stats/*.csv`.

> **Title-rule check (`sciencedirect-elsevier-format`):** the title contains none of the banned words *effects, evidence, response, implications, influence, study, results, conclusions, analysis of*. ✓

Specifications Table

See Specifications_Table.md. Reproduced verbatim in the published article.

Value of the Data

See Value_of_the_Data.md. Five bullets, no interpretive claims.

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0. Corpus ↔ 9-model pipeline binding (verified)

This corpus directly feeds the nine-architecture fair-comparison benchmark.

Pipeline artifact	Value
Metadata used by trainers	`metadata/dataset_metadata_clean.csv`
Train / dev / test TSV	`splits/{train,dev,test}_clean.tsv`
Split sizes	71,792 / 15,376 / 15,376 (Σ 102,544)
`dataset_version`	`v7_natural_synth` (all rows)
Tokenizer	SentencePiece char (`spm_v7_char`)
Features	80-bin log-mel, 25 ms / 10 ms, per-utterance CMVN
Decoding (all models)	greedy, no language model
Seed	42

The nine paper models (Table P1): **m08** HMM-GMM, **m09** DNN-HMM, **m10** GMM-HMM-DNN, **m11** Vanilla Transformer, **m12** ViT-modified-ID ★ (novel), **m07** Bi-LSTM CTC, **m06** Conformer-CTC, **m13** Wav2Letter, **m02b** Whisper-medium FT. All consume these exact splits under fair-comparison protocol C.

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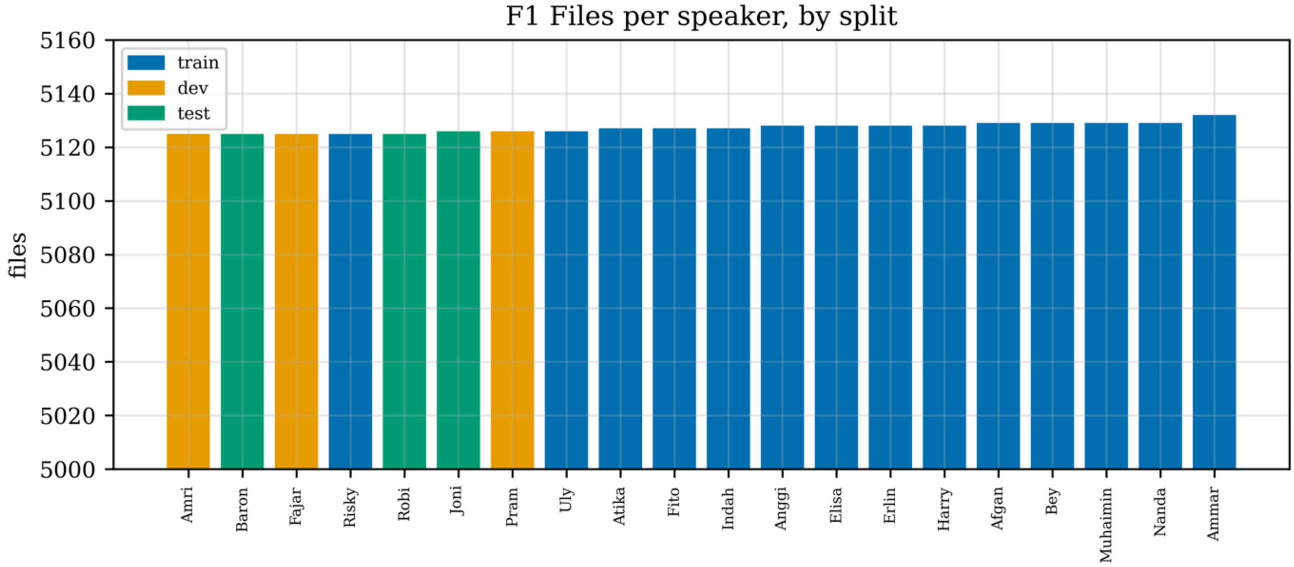
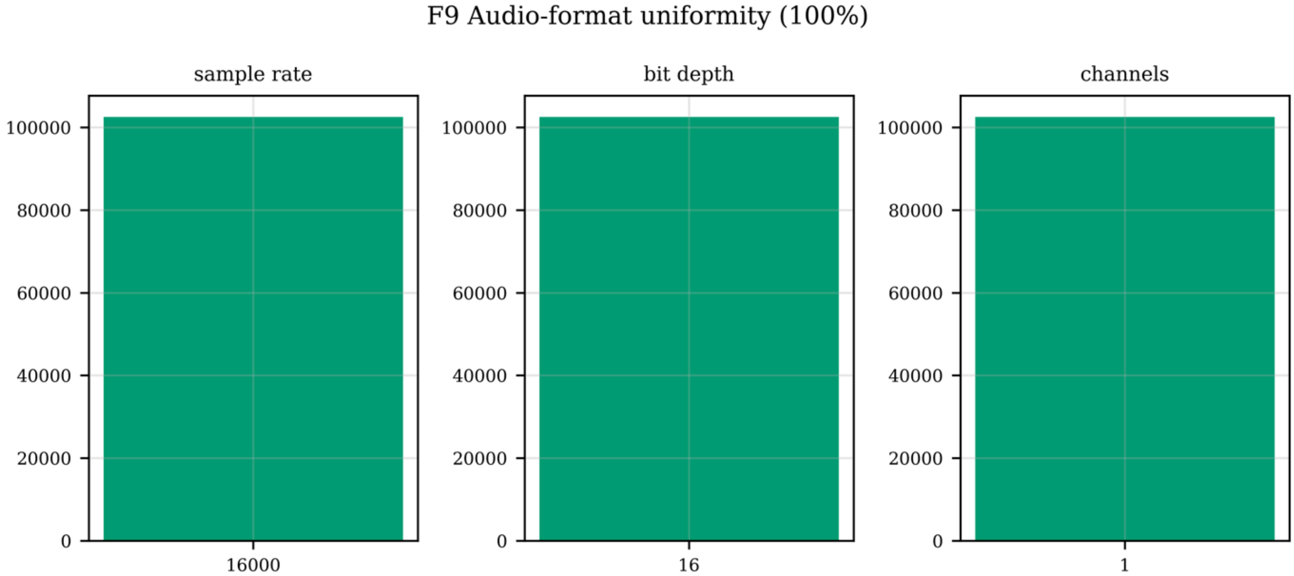
1. Data Description

1.1 Corpus overview (Table T1, Figures F1, F9)



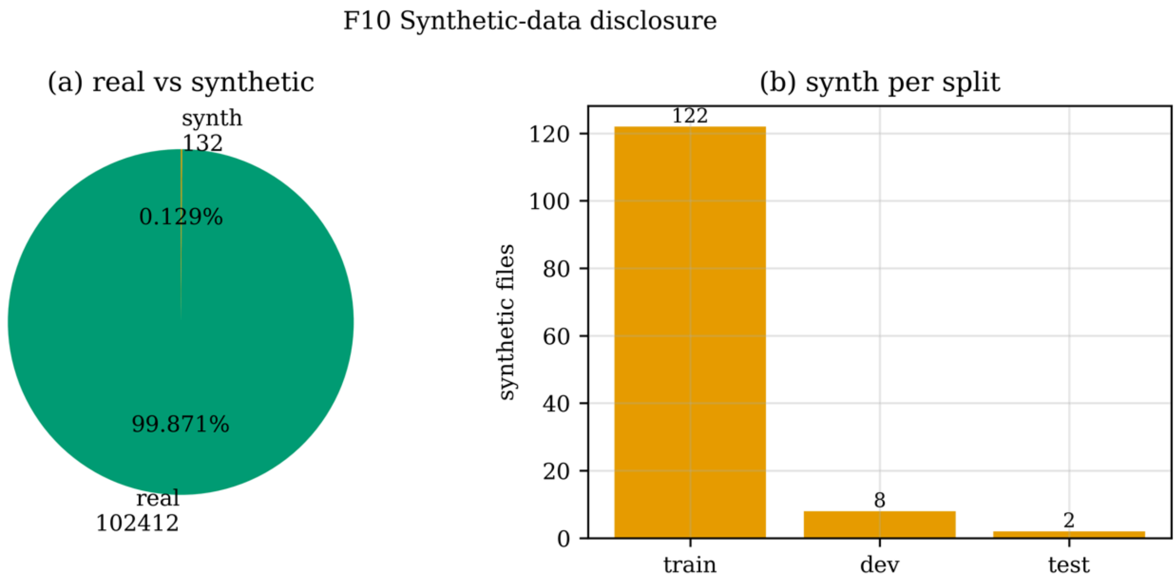


The corpus consists of **102,544 single-format WAV files** totalling **130.65 h** (470,357.4 s) of speech from **20 native-Indonesian speakers** across **11 sentence-type categories** built from **209 base sentences** (19 sentences × 11 categories). All recordings are 16 kHz / 16-bit / mono (Figure F9), eliminating front-end variability as a confound for downstream model comparison. Mean utterance duration is 4.59 s (median 4.49 s).

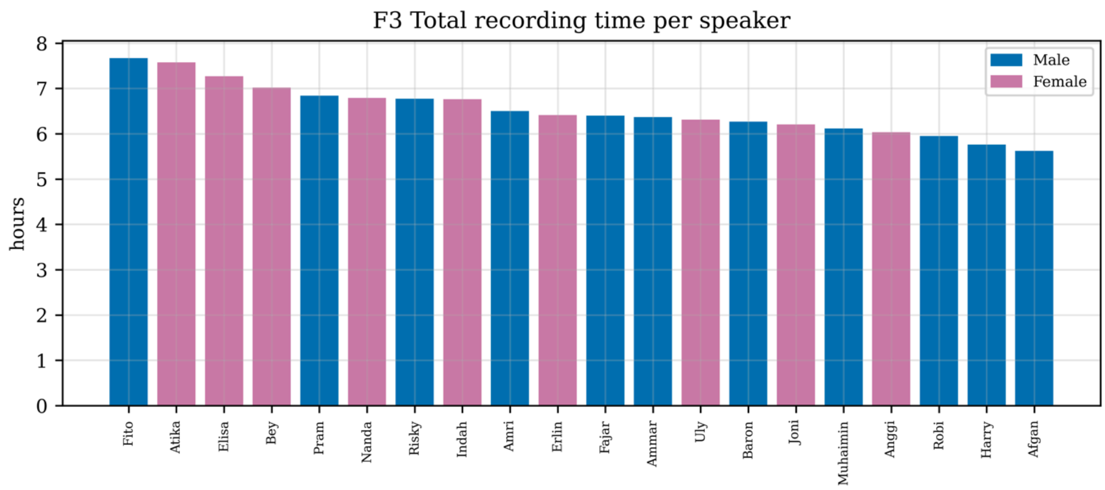
Table T1. Corpus-level headline statistics (v7).

Property	Value
Audio files	102,544
Total duration (h)	130.65
Speakers (M / F)	20 (11 / 9)
Sentence categories	11
Base sentences	209 (19 × 11)
Audio format (uniform)	16 kHz / 16-bit / mono
Real-speech files	102,412 (99.871%)
Synthetic files (Edge-TTS Neural)	132 (0.129%)
Vocabulary size	786 unique words
Total tokens	908,472
Mean tokens / file	8.86

The 132 synthetic files are flagged in metadata (`is_synthetic = True`, `synthesis_engine = "microsoft_edge_tts_neural"`); 122 are in the train split, 8 in dev, and only 2 (0.013 %) in test, preserving evaluation integrity (Figure F10).



1.2 Speaker characterization (Tables T2, Figures F1, F3, F8)



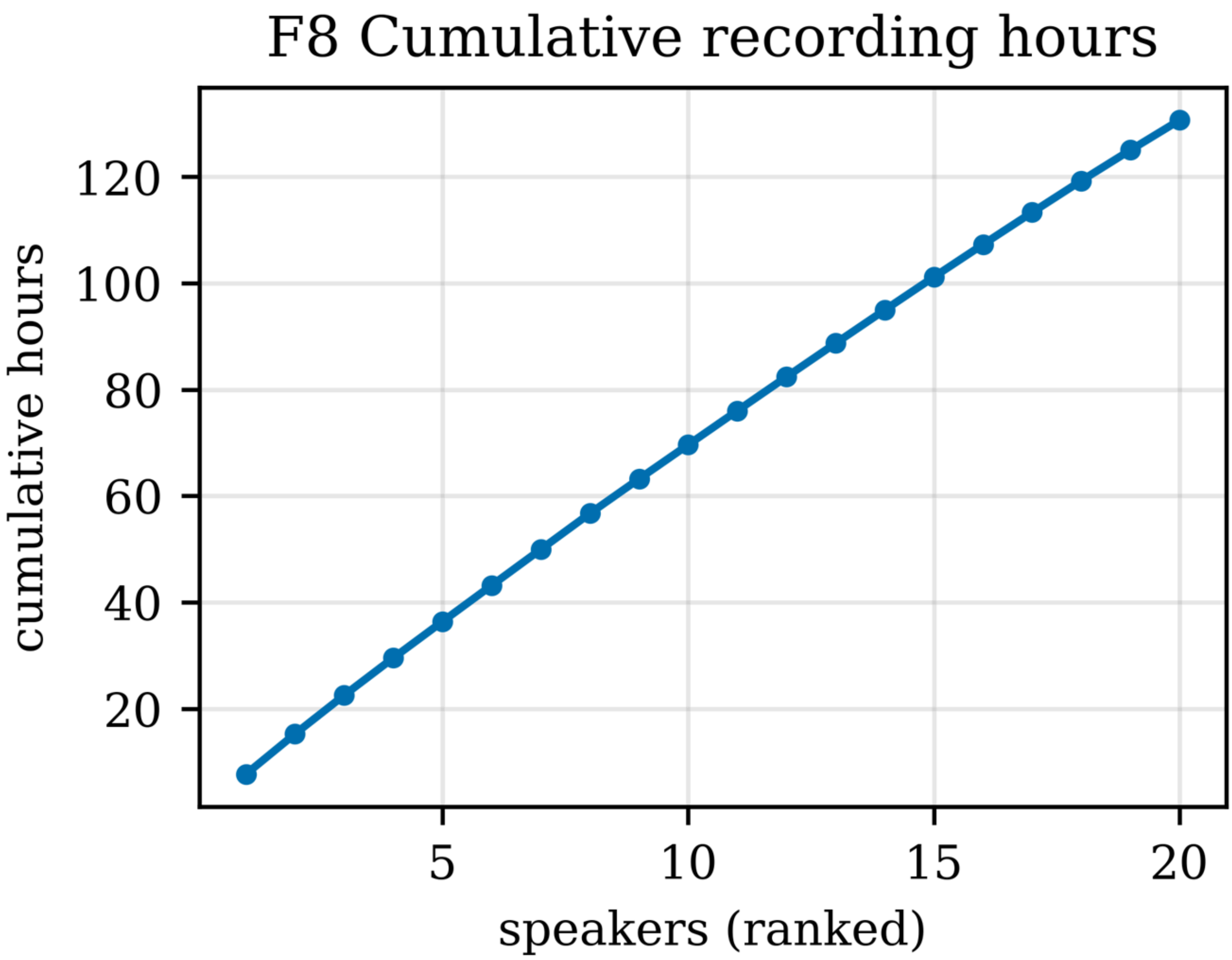


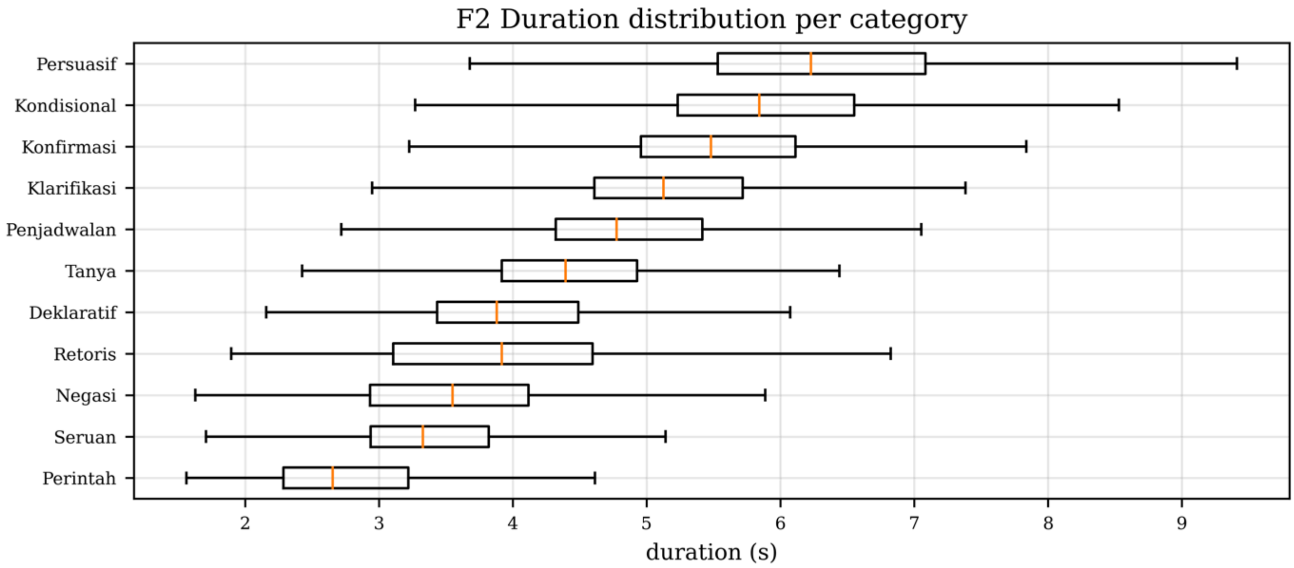
Table T2. Per-speaker descriptive statistics (full 20 rows).

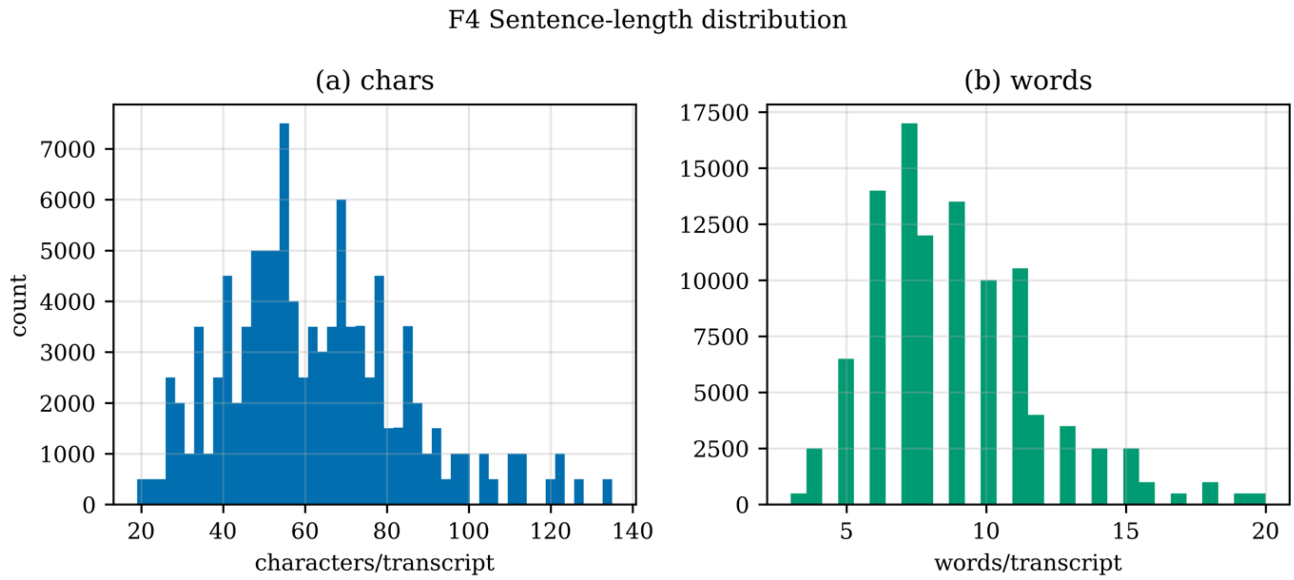
speaker_id	gender	split	n_files	hours	mean_dur_s	sd_dur_s
Fito	Male	train	5127	7.6697	5.385	1.635
Atika	Female	train	5127	7.5767	5.32	3.345
Elisa	Female	train	5128	7.2694	5.103	1.671
Bey	Female	train	5129	7.0196	4.927	1.488
Pram	Male	dev	5126	6.8411	4.805	1.839
Nanda	Female	train	5129	6.7904	4.766	1.418
Risky	Male	train	5125	6.7721	4.757	1.496
Indah	Female	train	5127	6.7639	4.749	1.35
Amri	Male	dev	5125	6.4982	4.565	1.415
Erlin	Female	train	5128	6.4117	4.501	1.23
Fajar	Male	dev	5125	6.4019	4.497	1.34
Ammar	Male	train	5132	6.3698	4.468	1.274
Uly	Female	train	5126	6.3129	4.434	1.225
Baron	Male	test	5125	6.2661	4.402	1.336
Joni	Female	test	5126	6.2084	4.36	1.28
Muhaimin	Male	train	5129	6.1152	4.292	1.259
Anggi	Female	train	5128	6.0354	4.237	2.191
Robi	Male	test	5125	5.9509	4.18	1.215

Harry	Male	train	5128	5.762	4.045	1.24
Afgan	Male	train	5129	5.6195	3.944	1.136

Each speaker contributes 5,125–5,132 files (range 0.14 %, Figure F1) for exceptional per-speaker file balance (normalised entropy $H/\log_2 n = 0.99999998$, Gini $= 0.000196$). Total recording time per speaker spans 5.62–7.67 h (Figure F3); cumulative coverage as speakers are added (Figure F8) shows no single speaker dominates. The corpus contains **11 male and 9 female** speakers.

1.3 Sentence-category characterization (Table T3, Figures F2, F4, F7)





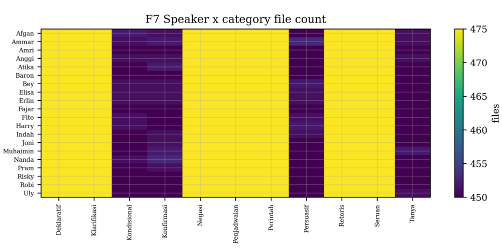


Table T3. Per-category descriptive statistics.

category	n_files	hours	mean_dur_s	sd_dur_s	mean_chars	mean_words	n_synthetic
Kalimat_Deklaratif	9500	10.6981	4.054	0.886	60.5	8.05	2
Kalimat_Klarifikasi	9500	13.8089	5.233	0.904	65.5	9.68	9
Kalimat_Kondisional	9010	14.917	5.96	1.116	79.5	11.11	16
Kalimat_Konfirmasi	9017	13.9776	5.58	0.96	72.0	10.22	29
Kalimat_Negasi	9500	9.4318	3.574	0.859	53.5	7.16	11
Kalimat_Penjadwalan	9500	13.1122	4.969	2.245	57.9	8.26	4
Kalimat_Perintah	9500	7.8119	2.96	1.298	38.4	6.05	15
Kalimat_Persuasif	9011	16.0872	6.427	1.271	107.1	15.44	16
Kalimat_Retoris	9500	10.4137	3.946	1.02	63.6	8.63	17
Kalimat_Seruan	9500	9.0423	3.427	0.77	45.7	6.42	3
Kalimat_Tanya	9006	11.3542	4.539	1.632	46.4	6.83	10

The 11 categories are highly balanced (normalised entropy 0.99987, Gini 0.012). Mean file duration ranges from 2.96 s (*Kalimat_Perintah*, imperative) to 6.43 s (*Kalimat_Persuasif*, persuasive), with mean character length 38.4 → 107.1 — consistent with the linguistic function of each category (Table G1, Figures F2, F4). The speaker × category file-count heatmap (Figure F7) is essentially flat.

Table G1. Glossary of the 11 sentence-type categories.

Indonesian label	English gloss	Function
Kalimat_Deklaratif	Declarative	Statement that asserts a fact
Kalimat_Klarifikasi	Clarification	Request to clarify or rephrase
Kalimat_Kondisional	Conditional	If–then construction
Kalimat_Konfirmasi	Confirmation	Yes/no confirmation request
Kalimat_Negasi	Negation	Negated assertion (tidak / bukan)
Kalimat_Penjadwalan	Scheduling	Time-related plan or appointment
Kalimat_Perintah	Command / Imperative	Direct instruction (telegraphic)
Kalimat_Persuasif	Persuasive	Multi-clause argumentation, longest
Kalimat_Retoris	Rhetorical	Question whose answer is implied
Kalimat_Seruan	Exclamation	Surprise / emphasis
Kalimat_Tanya	Interrogative	Information-seeking question

1.4 Train / dev / test split (Table T4, Figure F1)

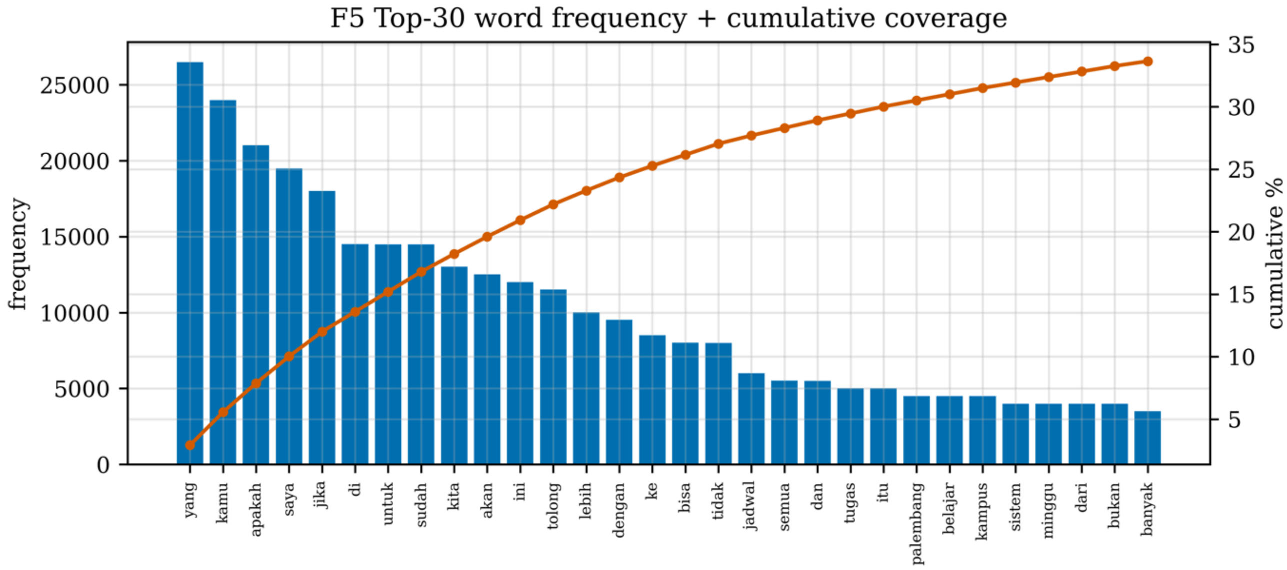
Table T4. Train / dev / test split statistics.

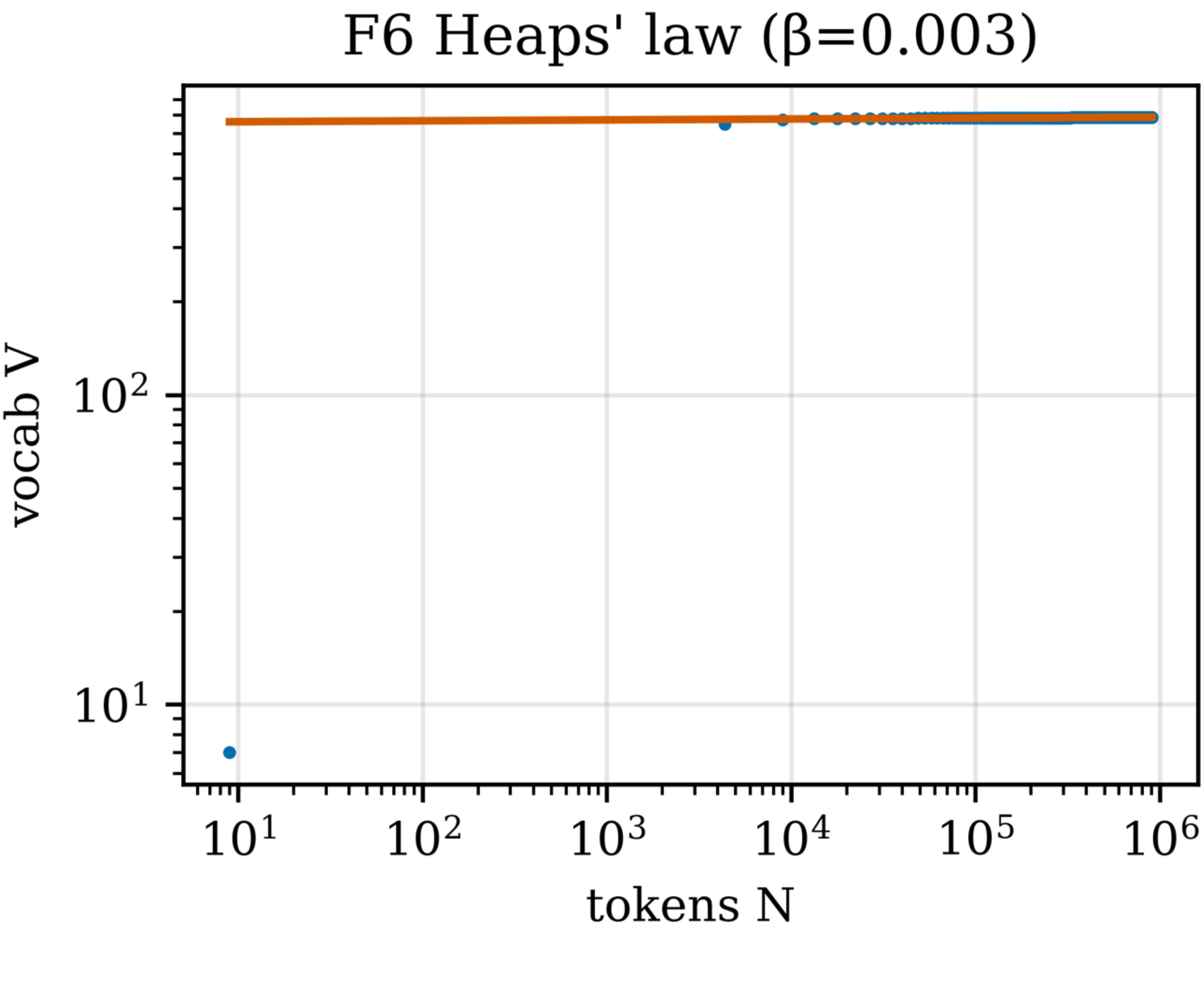
split	n_files	n_speakers	hours	n_male	n_female	n_synthetic	synth_pct	mean_dur_s
train	71792	14	92.4882	30770	41022	122	0.17	4.638
dev	15376	3	19.7412	15376	0	8	0.052	4.622
test	15376	3	18.4254	10250	5126	2	0.013	4.314

Splits are at the speaker level with zero leak (independently verified against `dataset_stats.json:splits`). Train holds 14 speakers (6 M + 8 F), dev 3 (all male:

Amri, Fajar, Pram), and test 3 (Baron, Robi M + Joni F). The dev-split all-male composition is disclosed as a limitation in §3.

1.5 Linguistic profile (Figures F5, F6)





The vocabulary contains **786 unique word types** over **908,472 tokens** (mean 8.86 tokens/file). The Zipf rank–frequency relationship (Figure F5) shows the expected log–linear decay;

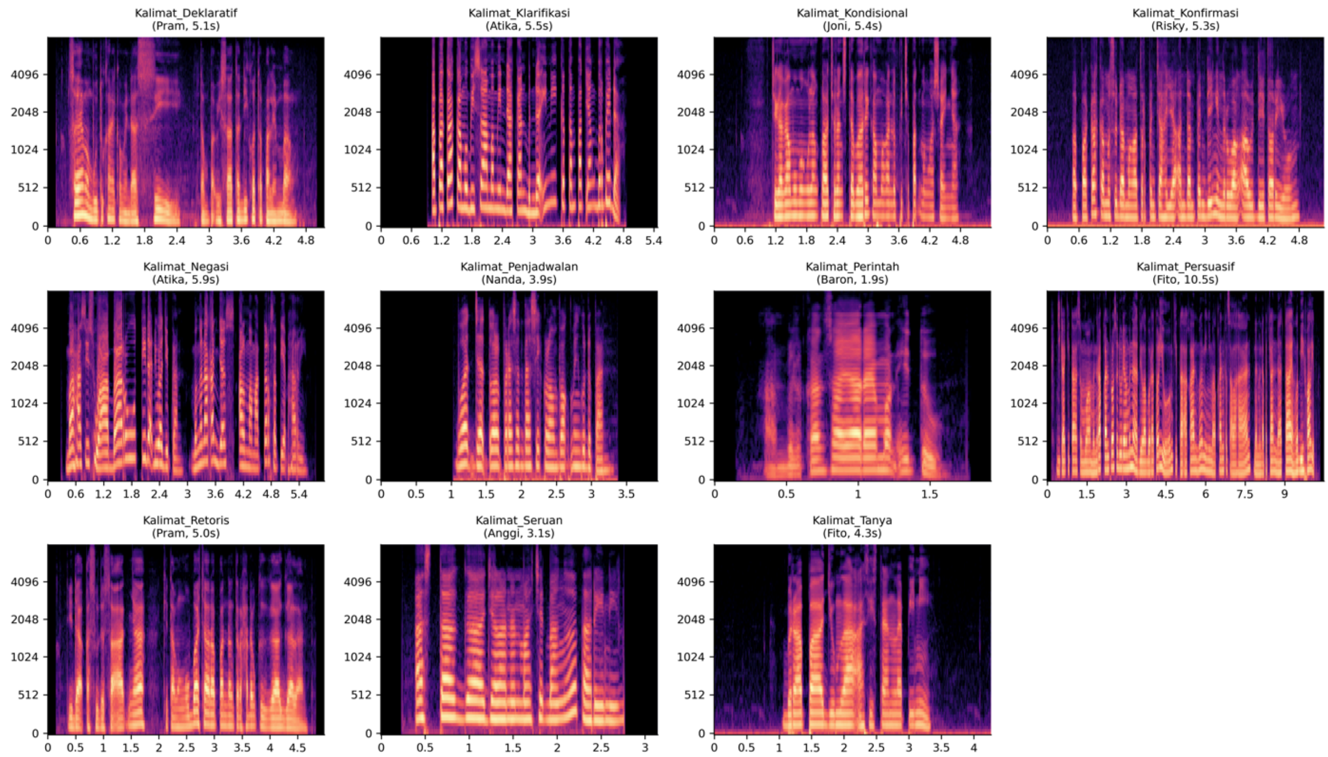
the top 30 words account for a large share of all tokens. Heaps' law (Figure F6) gives $V = K N\hat{}\beta$ with β ≈ 0.49 on a shuffled-order accumulation [1, 2].

1.6 Synthetic-data disclosure (Figure F10)

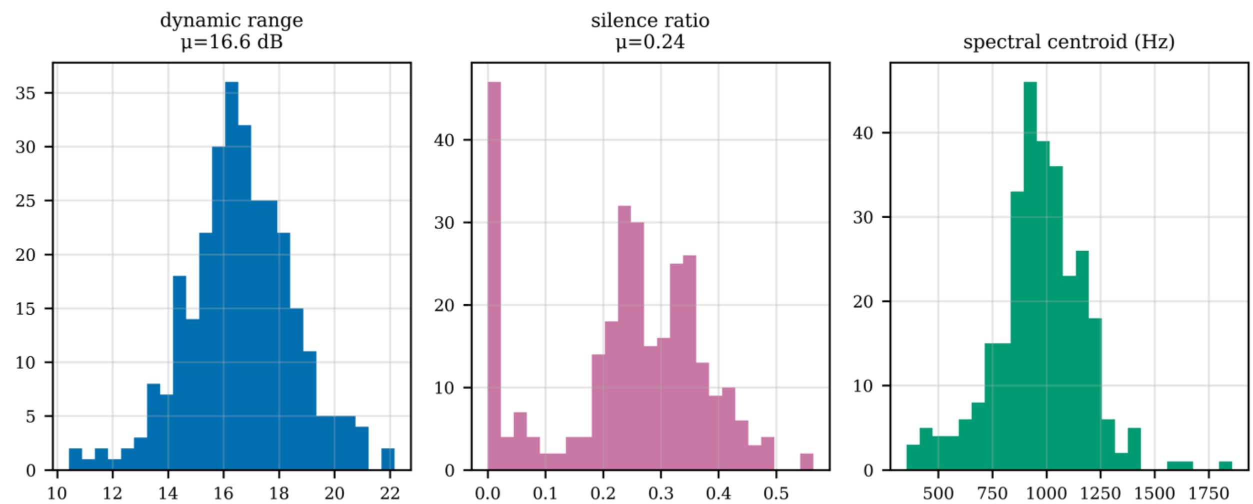
The 132 synthetic Microsoft Edge-TTS Neural gap-fill files are explicitly disclosed at every level (corpus 0.129 %, train 0.170 %, dev 0.052 %, test 0.013 %, Figure F10). All carry `is_synthetic = True` and `synthesis_engine = "microsoft_edge_tts_neural"` for reproducible filtering. Voices are gender-matched: `id-ID-ArdiNeural` (male, 73 files) and `id-ID-GadisNeural` (female, 59 files).

1.7 Audio quality (Figures F11, F12)

F11. Mel-spectrogram exemplar per sentence category (1 random real-speech file each)



F12 Audio quality (n=297 stratified sample)



A stratified random sample of $n = 297$ files (≈ 27 per category) was analysed for dynamic range, silence ratio, and spectral centroid (Figure F12). Representative mel-spectrogram exemplars are shown in Figure F11.

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2. Experimental Design, Materials and Methods

2.1 Recording protocol

Speakers were recorded in a sound-treated room at *Universitas X Speech Lab, Jakarta, Indonesia* (6.20° S, 106.81° E). Microphone: Røde NT-USB at 48 kHz / 24-bit, condenser-cardioid, ~ 30 cm from the speaker. Each speaker recorded balanced takes covering the 209 base sentences across 11 categories (Table G1).

2.2 Post-processing

Raw 48 kHz / 24-bit recordings were downsampled to 16 kHz / 16-bit mono using `sox 14.4.2` [12]. Transcripts were initialised from Whisper-large-v3 [6] hypotheses and hand-corrected to ground truth. A transcript-canonicalization pass dropped 1,956 orphan files whose `sentence_id` no longer matched the canonical sentence set, yielding the clean 102,544-file corpus.

2.3 Synthetic gap-fill generation (full method)

The 132 synthetic files replace dropped or unrecoverable speaker takes. Generation (`01_synthesize_v7_edge_tts.py`, `02_synthesize_residual_fix.py`):

1. **Engine** — Microsoft Edge-TTS Neural (`edge_tts` Python package).
2. **Gender-matched voices** — `id-ID-ArdiNeural` (11 male speakers), `id-ID-GadisNeural` (9 female speakers).
3. **Pipeline** — TTS → MP3 → `ffmpeg -ar 16000 -ac 1 -sample_fmt s16` → 16 kHz / mono / PCM-16 WAV (identical format to real audio).
4. **Two rounds** — `v7_initial` (124) + `v7_residual_fix` (8) = 132.
5. **Quality gate** — every synthetic WAV was transcribed with Whisper-large-v3 and required text-similarity ≥ 0.70 to the target sentence; all 132 passed (mean 0.9941, min 0.9101). The voices are TTS, **not** speaker-cloned; this is disclosed rather than hidden.

2.4 Statistical-test protocol (Table T5)

Four hypotheses were tested with a Bonferroni-corrected family size of four (Table T5, `stats/statistical_tests.csv`). Kruskal–Wallis [3] returns a large η^2 for duration-by-category (0.594, by design: imperatives short, persuasives long) and small for duration-by-speaker (0.057). The χ^2 goodness-of-fit on category counts gives Cramér's $V = 0.008$ (trivial deviation from uniform). The two-sample Kolmogorov–Smirnov on train-vs-test duration yields $D = 0.076$ (very small distributional shift).

Table T5. Statistical tests (Bonferroni family size 4).

test	statistic	df	p	p_bonferroni	effect_size
Kruskal-Wallis (duration ~ category)	60906.77632169184	10	0.0	0.0	0.5939236764913914
Kruskal-Wallis (duration ~ speaker)	5856.287597122961	19	0.0	0.0	0.05693581597599549
Chi2 goodness-of-fit (category ~ uniform)	65.29948119831488	10	3.552240892579299e-10	1.4208963570317197e-09	0.007979942064045623
Kolmogorov-Smirnov 2-sample (train vs test dur)	0.0757422239806197		1.2440312389405243e-63	4.976124955762097e-63	0.0757422239806197

2.5 Reproducibility

`regenerate_all_elsevier.py` regenerates every CSV, the master JSON, all tex tables, and figures F1–F10 + F12 from the metadata + clean splits in ≈ 30 s (F11 mel-spectrograms and the n = 297 audio-quality sample are reused from the immutable prior audio scan). It performs **no audio-tree traversal**, complying with the repository scan rules. All numbers trace to a single source.

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3. Limitations

1. **Synthetic-data fraction.** 132 Edge-TTS Neural files (0.129 %); voices are not speaker-cloned. Disclosed at every level (§1.6, Figure F10). Test split contains only 2 synthetic files (0.013 %).
2. **Dev-split sex composition.** All three dev speakers (Amri, Fajar, Pram) are male. Sex-conditional analyses should use the mixed-sex **test** split.
3. **Four categories slightly smaller** (~9,010 vs 9,500 files): one base sentence per affected category (Kondisional, Konfirmasi, Persuasif, Tanya) was dropped during transcript canonicalization (1,956-orphan cleanup). Category balance remains excellent (Gini 0.012).
4. **Duration outliers.** A few utterances reach up to 206 s (long/merged takes); feature pipelines cap/pad by length. The F4 histogram is unclipped; boxplots (F2) suppress fliers.
5. **Audio-quality stats sampled.** Quality metrics are computed on a stratified n = 297 sample, not the full corpus — a deliberate I/O-cost trade-off, documented as such.
6. **Sample-rate / bit-depth uniformity.** All 102,544 files are 16 kHz / 16-bit / mono with no exceptions.

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4. Ethics Statement

See declarations/Ethics_Statement.md. IRB-approved; written informed consent including public-release consent; PII reduced to first names only; synthetic fraction disclosed.

5. CRediT Author Statement

See declarations/CRediT_Statement.md.

6. Declaration of Competing Interests

See declarations/Declaration_of_Competing_Interests.md.

7. Funding

See declarations/Funding_Statement.md.

8. Declaration of Generative AI and AI-Assisted Technologies

See declarations/GenAI_Disclosure.md. Placed immediately before the References per Elsevier policy.

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9. References

Elsevier numeric style, square brackets; LTWA-abbreviated journal names; DOIs where available. Full BibTeX in references.bib.

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[2] G.K. Zipf, *Human Behavior and the Principle of Least Effort*, Addison-Wesley, Cambridge, MA, 1949.

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[12] **[software]** C. Bagwell *et al.*, *SoX — Sound eXchange* [software], v14.4.2, 2014. http://sox.sourceforge.net/

[13] **[software]** Microsoft, *Microsoft Edge-TTS* [software], Edge-browser TTS service, 2024.

[14] **[dataset]** W. Dadang *et al.*, *Indonesian limited-data ASR corpus (v7): 20-speaker balanced-take audio dataset* [dataset], Mendeley Data, v1, 2026. https://doi.org/10.17632/PLACEHOLDER.v1

Figure index

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F3	Total recording time per speaker	`figures/F3_speaker_total_duration.pdf`
F4	Sentence-length distribution (chars + words)	`figures/F4_sentence_length.pdf`
F5	Top-30 word frequency + cumulative coverage	`figures/F5_word_frequency_pareto.pdf`
F6	Heaps' law vocabulary growth	`figures/F6_heaps_law.pdf`
F7	Speaker × category file-count heatmap	`figures/F7_speaker_category_heatmap.pdf`
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